Mapping Natural-Disaster Damage from Satellite Imagery

Cross-disaster transfer on xBD: pretrained versus from-scratch change-detection CNNs

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Deep Learning Final Project

Abstract. This project addresses two questions about automated building-damage classification on xBD: whether performance transfers from a set of disaster types seen in training to a disaster type withheld entirely, and whether ImageNet pretraining improves that transfer relative to a comparably designed encoder trained from scratch. Two siamese change-detection CNNs that differ only in the encoder, one an ImageNet-pretrained ResNet-18, one a compact encoder trained from scratch, are trained on 14 non-fire disaster events under identical patches, splits and class weights, then evaluated both in-distribution and zero-shot on five wildfire events held out entirely. The pretrained model reaches 0.808 accuracy and 0.650 macro-F1 in distribution, falling to 0.774 accuracy and 0.429 macro-F1 on wildfire, a 33.9% relative loss in macro-F1. This aggregate drop does not indicate a general representation failure: destroyed scores higher out of domain than in it (F1 0.758 versus 0.687), no-damage precision reaches 0.966, and threshold-free ROC-AUC holds at 0.814. The degradation is confined to the two intermediate damage classes, which make up 19.1% of training patches but only 1.7% of wildfire patches, indicating that the learned visual features transfer across disaster types while the decision calibration, tuned to the training-domain class prior, does not. Comparing the two encoders, pretraining outperforms training from scratch on every split and every metric, with the largest margin precisely where it matters most: +0.143 accuracy and +0.052 macro-F1 on the wildfire set. A third analysis joins 185,045 labelled buildings to census block groups and finds that an apparent association between median income and damage ($\beta = +0.225$, $p = 0.0009$) disappears once event fixed effects are added ($\beta = +0.057$, $p = 0.44$): the association is explained by which disaster struck which place, not by income differences within a disaster. Recovery capacity tells a different story: pricing the same damage in dollars against HAZUS loss ratios, validated against NFIP, CAL FIRE and FEMA records, exposes a steep income gradient, with 61% of the poorest income quartile's flood loss covered by neither insurance nor federal aid against 34% in the richest, and auditing the two Part 2 models' own predictions shows that an automated relief rule would misdirect $30–78M of every $1B on seen disaster types and $112–200M on wildfire, the type the models never trained on.

1 Introduction

Automated damage maps are operationally useful only if they generalise to disaster types for which no labelled data yet exists. In-distribution accuracy does not measure this: a model can perform well on held-out data drawn from the same disaster types it was trained on while failing on a disaster type it has never encountered. This report addresses two related questions. First, how much of a change-detection model's performance transfers when it is evaluated on a disaster type withheld entirely from training. Second, whether ImageNet pretraining improves that transfer relative to a comparably designed encoder trained from scratch on the same data, since pretraining is a design choice available at no additional labelling cost.

We study these questions on xBD [1], the dataset behind the xView2 challenge [9], which pairs pre- and post-event high-resolution satellite tiles with building footprints and a four-level damage scale. Prior work on this data fuses the two time steps inside the network [2]; we instead keep the comparison explicit in the classification head, for the reason given in Section 5.2. The task is framed as change detection at building level: given two co-registered crops of the same structure, one before and one after the event, predict no-damage, minor-damage, major-damage or destroyed.

The experimental design avoids a random split. Five wildfire events are withheld from training entirely and used as a zero-shot test set, while the official xView2 evaluation directories provide an in-distribution reference measured on the same trained model. Both encoders, pretrained and from-scratch, are evaluated identically on both sets, so the results address the transfer question and the pretraining question together rather than in isolation.

2 Data

2.1 Source and access

We use the full xBD release: 11,034 scenes over 19 events, each contributing a 1024 × 1024 RGB pre-disaster tile, the matching post-disaster tile, and a GeoJSON label file. That is 22,068 images and 425,368 annotated building polygons on the post-disaster side, obtained through a public mirror of the xView2 challenge release.[1]

Two properties of the source were verified rather than assumed. First, the tiles are co-registered: pixel (x, y) denotes the same ground location before and after the event, which is what makes cropping an identical bounding box from both tiles valid; without this property the siamese design would be comparing different places. Second, the format is uniform: all 800 files in a stratified quality-control sample decode as 1024 × 1024 RGB PNG with no read errors, so no resolution standardisation or band alignment is required. Damage labels are recorded only in the post-disaster GeoJSON, under `features.xy[].properties.subtype`; the same file carries `features.lng_lat`, giving per-building WGS84 coordinates, so the geo-referencing for Part 3 is already present.

[1] Kaggle dataset `qianlanzz/xbd-dataset`, layout `xbd/{hold,test,tier1,tier3}/{images,labels}`.

Table 1: Split design and sizes. Only `train`/`val` depend on the random seed; both test sets are defined by the data itself.

Split	Rule	Scenes	Polygons
`train`	tier1+3, non-fire	3,135	225,506
`val`	tier1+3, non-fire	392	30,794
`test_id`	test+hold, non-fire	1,135	91,755
`test_ood`	wildfire, all	6,372	77,313
Total		11,034	425,368

2.2 Split design

The split is scene-level and was produced by a dedicated indexing script rather than inside a training notebook, so every model in the project trains on an identical partition (Table 1).

Two aspects of the split design merit comment. First, `test_id` is drawn from the official `test` and `hold` directories, which were never part of any training pool and, unlike `val`, were never used for tuning, so they provide an honest reference for the headline number. Second, assignment happens at scene level, which guarantees that the pre- and post-disaster image of a location cannot separate and that no building can appear in two splits.

The five held-out events are `pinery-bushfire`, `portugal-wildfire`, `santa-rosa-wildfire`, `socal-fire` and `woolsey-fire`. An automated check confirms zero wildfire scenes in `train` or `val`, and zero event overlap between the training pool and `test_ood`.

3 Exploratory analysis

3.1 Class imbalance is severe and split-dependent

no-damage accounts for 72.0% of training polygons and the rarest class is 11.6× smaller (Fig. 1). More importantly the composition is not stable across splits: `test_ood` is 80.2 / 0.9 / 0.9 / 14.7% against 72.0 / 9.7 / 8.4 / 6.2% in `train`.

Plain accuracy is therefore a poor decision metric: always predicting no-damage scores 0.72 on the training distribution while producing a damage map that never flags damage. We report macro-F1 and per-class precision and recall as the decision metrics, use inverse-frequency class weights computed on `train` only (0.330/2.379/2.910/4.919), and key checkpointing and early stopping on validation macro-F1. Weighting was preferred to resampling: oversampling would repeat the same few destroyed buildings many times per epoch and invite memorisation, while undersampling would discard most of the majority signal.

3.2 Wildfire damage is bimodal

Averaged over events, the non-fire domain is 70.2 / 9.9 / 7.2 / 6.5% across the four classes while the wildfire domain is 78.8 / 1.2 / 1.1 / 14.8%. Fire either leaves a building intact or destroys it; intermediate states barely occur. Figure 2 shows this per event, and also how much the non-fire events differ among themselves: `mexico-earthquake` is 99% undamaged, `hurricane-matthew` is 51% minor-damage, `hurricane-harvey` is 35% major-damage.

This bimodality operationalises the project's central hypothesis: a model that implicitly learns the class prior of the non-fire training events will be miscalibrated on fire even if its visual features transfer perfectly. It also specifies what to look for in the out-of-domain predictions: a pure prior shift should produce errors with a consistent direction, whereas a failure of the learned features should produce diffuse, undirected errors.

3.3 Scene occupancy and patch size

Empty scenes are common and strongly event-dependent (Fig. 3): 9.8% of training scenes contain no buildings, against 54.1% of wildfire scenes, whose median building count is zero, because the tier3 fire events cover large stretches of forest and scrubland. An empty scene yields no samples but still costs a full 1024 × 1024 decode, so we sample only scenes with at least five usable buildings. This is a sampling decision, not a cleaning one: nothing is deleted, and the identical rule applies to all four splits so it cannot tilt the comparison.

Buildings are small in absolute pixels: the median longest bounding-box side is 35 px and the 90th percentile 64 px, with a thin tail to 349 px at the 99th. A 64×64 patch therefore covers roughly 90% of footprints at close to native resolution; larger mostly upsamples empty ground at quadratic cost, smaller destroys the roof texture separating minor from major damage. We keep a 10 px margin around each footprint, leaving a 44 × 44 px window for the building. Damage is often more visible around a building than on it (debris, scorch marks, standing water), but the margin is kept modest deliberately, for reasons Section 7 makes concrete.

3.4 A confound in the acquisition geometry

Ground sample distance and off-nadir angle are similar across the two domains (2.21 ± 0.52 against 2.12 ± 0.31 m, and 28.5 ± 9.9 against 25.6 ± 6.0 degrees), but sun elevation is not: 64.4 ± 7.5 degrees for non-fire against 55.0 ± 15.9 for wildfire, with a fire mode near 35° that has no counterpart in training. Low sun means long shadows and a materially different appearance for the same structure.

Limitation: confounded acquisition geometry. These spreads overstate their own precision, because every scene within an event shares a single acquisition. Statistically the effective sample size is 19 events, not 11,034 scenes, and acquisition geometry is fully confounded with event identity: wildfire and the specific captures that happen to be wildfire cannot be sepa-

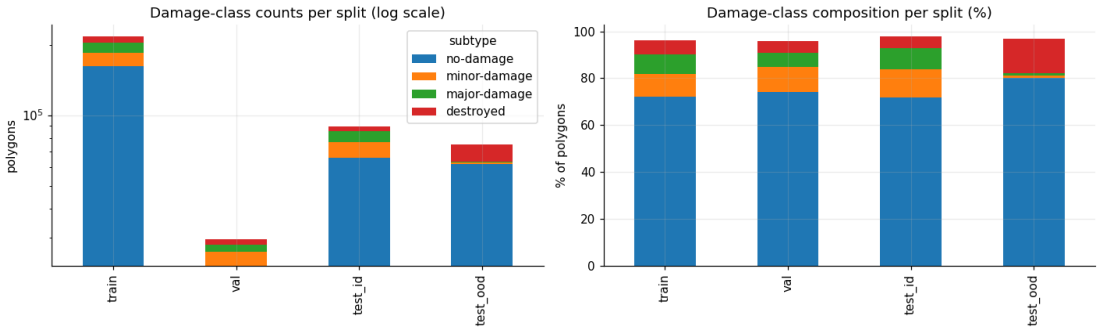


Figure 1: Damage-class distribution per split. Left: counts on a log scale. Right: composition in per cent. The `test_ood` bar is qualitatively different from the other three, with the intermediate classes almost vanishing and destroyed more than doubling.

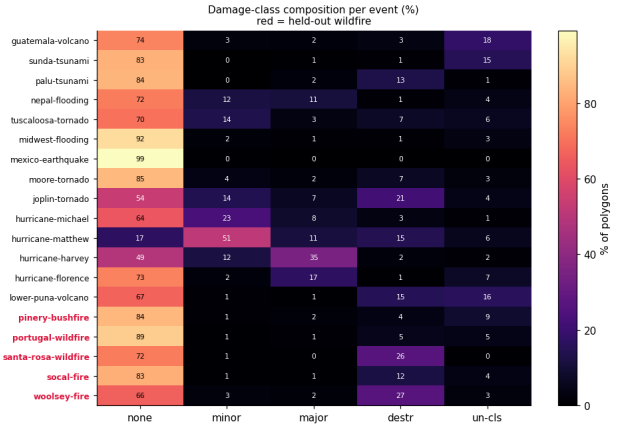


Figure 2: Damage-class composition per event, in per cent of that event's polygons. Held-out wildfire events in red. The fire rows share a signature that no non-fire event reproduces: near-zero minor/major with elevated destroyed.

rated on this data. Part of the transfer gap reported below may therefore be attributable to shadow length rather than to fire damage itself, and we cannot bound how much.

3.5 What the classes look like

Figure 4 contrasts the two domains directly. In the non-fire domain the post-disaster image usually retains the structure with altered geometry: torn roofs, debris, standing water. Wildfire damage instead collapses a building to a uniform grey scar with the footprint faintly visible, while undamaged buildings often sit in surroundings that changed completely (scorched vegetation) with the roof untouched.

That last observation generates a falsifiable prediction, made before any model was trained: a change-detection model learns "the scene changed" ⇒ "the building is damaged", so under wildfire it should over-flag no-damage. Section 6 reports the outcome.

4 Data cleaning

Nothing is removed silently. Every rule states its threshold, its cost and its per-split removal rate, so a rule that quietly favours one split can be caught. Table 2 is the ledger.

Table 2: Cleaning ledger. Removal is reported against the scope each rule operates on. The 14 occlusion scenes and the 28 near-duplicate pairs were flagged for inspection, not deleted.

Rule	Scope	Removed
Missing / unpaired	11,034 scenes	0
Corrupted images	800 files	0
Unreadable labels	11,034 scenes	0
Smoke / cloud	683 scenes	0
un-classified	425,368 polys	14,011
Degenerate polygons	425,368 polys	0
Side < 8 px	411,357 polys	14,458
Near-duplicates	400 scenes	0
Usable after cleaning (93.3% of 425,368)		**396,899**

The two rules that cost data are the un-classified subtype (3.29% of all polygons) and sub-8-pixel footprints (3.51% of the labelled remainder). Both were checked per split. The un-classified rate is 3.67 / 4.08 / 2.09 / 3.30% across train / val / test_id / test_ood, and the tiny-footprint rate is 3.77 / 3.30 / 2.99 / 3.49%. Neither is concentrated in `test_ood`, so neither flatters the transfer result.

4.1 The occlusion question: a negative result

An open data-quality question for this project was whether images obscured by smoke should be removed. xBD carries no occlusion label: the metadata block contains only capture geometry. The closest available proxy is the un-classified subtype, which annotators used when a building could not be assessed.

We ranked scenes by that proxy and tested it against four whole-tile statistics, chosen because smoke and thin cloud on 3-band optical imagery are bright, low-contrast, low-saturation and edge-poor. The sample is enriched deliberately, a 400-scene stratified quality-control sample plus the 300 scenes with the highest un-classified rate, so the diagnostic has range on the proxy rather than relying on chance coverage. Over the 505 tiles with a measurable rate the Spearman correlations are -0.366 (edge density), $+0.116$ (brightness), -0.095 (saturation) and -0.035 (contrast). Edge density is

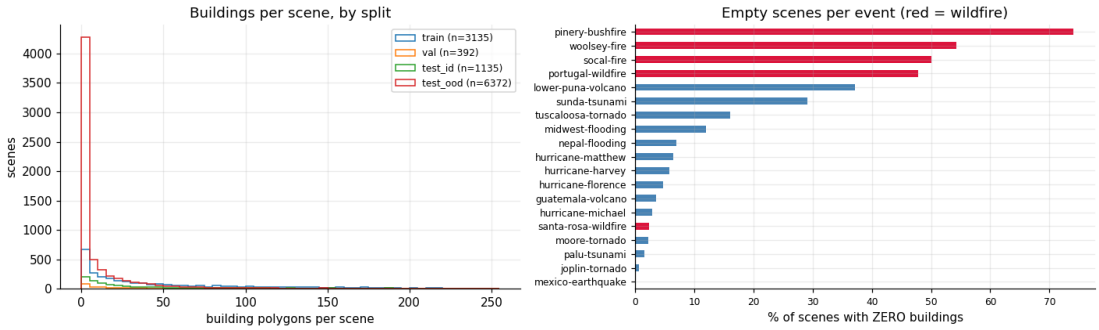
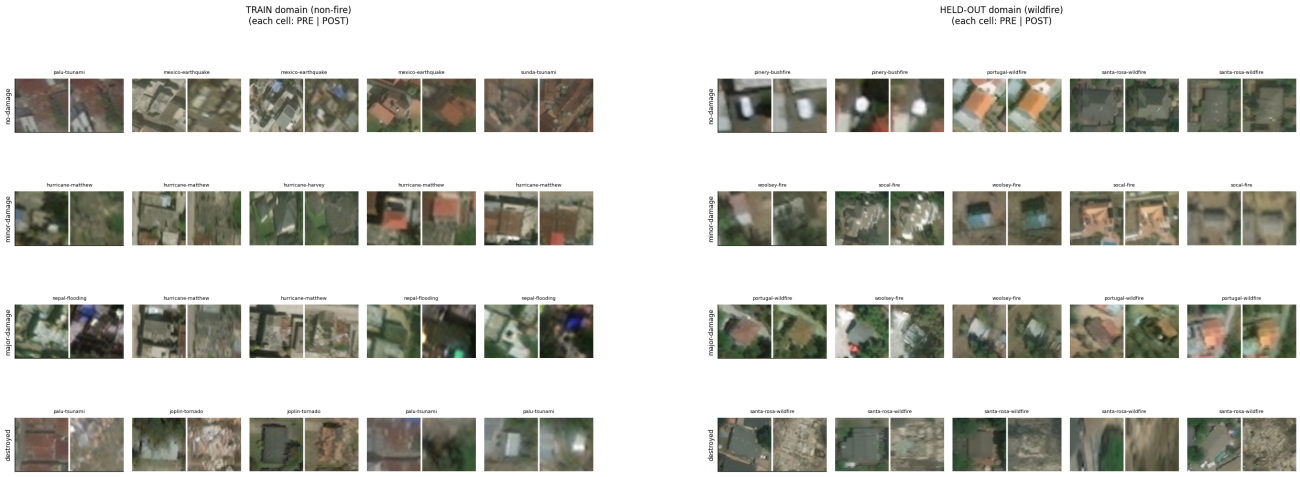


Figure 3: Left: distribution of building count per scene by split. Right: share of scenes with zero buildings, per event (wildfire in red). Four of the five wildfire events are close to or above half empty; pooled over all wildfire scenes the figure is 54.1%.



(a) Training domain (non-fire). **(b)** Held-out domain (wildfire).

Figure 4: Representative building patches, one row per damage class. Each cell shows the pre-disaster crop on the left and the post-disaster crop on the right.

the only one with any strength, explaining about 13% of the rank variance: real, but nowhere near a usable filter. Brightness, the statistic smoke should raise most, is essentially uncorrelated. Neither signal can be trusted alone, and that absence of correlation is itself the finding.

Result, and why nothing was removed. Flagging the top 2% of a composite occlusion score returns 14 scenes: 13 `sunda-tsunami` and 1 `lower-puna-volcano`, i.e. cloud over non-fire events, and none is a wildfire scene. By split they are 11 `train` and 3 `val`, with zero in either test set. These scenes were flagged but not removed, for three reasons: there is no ground truth against which to validate a threshold, nothing flagged touches a reported metric, and occlusion is part of the deployment problem in any case, since a system running on real post-disaster imagery will encounter cloud and smoke. The result that matters here is negative: the original concern, smoke concentrated over wildfire, is not what the diagnostic finds.

4.2 Leakage checks

A perceptual-hash duplicate test over the 400-scene quality-control sample returned 28 near-duplicate pairs, 19 crossing a split boundary. This was investigated rather than waved through: every cross-split pair traces back to a handful of low-texture tiles colliding with several different events on different continents. One `portugal-wildfire` tile alone matches tiles from `nepal-flooding`, `hurricane-harvey`, `sunda-tsunami`, `palu-tsunami` and `moore-tornado` at once, and one tile each from the last three of those is pairwise identical at Hamming distance 0. A single wildfire tile cannot be five unrelated disasters on other continents. The cause is hash degeneracy: a 64-bit difference hash of a low-texture tile (open water, unbroken forest, solid cloud) collapses toward a constant, so flat tiles collide regardless of content. We find no evidence of leakage, consistent with the scene-level split design.

5 Method

5.1 Patch construction

One sample is two $64 \times 64 \times 3$ `uint8` patches cropped from the same pixel bounding box in the pre- and post-disaster tile, plus one integer label. Because the tiles are already co-registered no warping is required; preprocessing is crop, resize, scale to $[0, 1]$ and ImageNet normalisation. Augmentation (flips, 90-degree rotations, brightness jitter) is applied identically to

Table 3: Extracted patch pairs per split and class. Note the support for minor and major in `test_ood`: every per-class number we report for those two classes rests on these counts.

	none	minor	major	destr	**total**
`train`	55,284	7,661	6,265	3,706	72,916
`val`	15,701	2,571	1,164	1,010	20,446
`test_id`	20,500	2,840	2,512	1,421	27,273
`test_ood`	12,056	**116**	**138**	2,317	14,627
Total					**135,262**

both images of a pair, since independent jitter per branch would manufacture a difference where none exists, which is precisely the signal the model is meant to measure.

Scenes were sampled under a compute budget, stratified by event and restricted to scenes with at least five usable buildings, yielding 135,262 patch pairs from 1,698 scenes (Table 3). Patches are written to memory-mapped arrays rather than individual image files, which keeps the working set on disk and the loader fast.

5.2 Architecture

The task is a comparison, not the classification of a single image. Using only the post-disaster patch discards the reference, while stacking pre and post into a six-channel input forces the first convolution to be relearned, throwing away the pretrained weights exactly where they transfer best. Both models therefore use a siamese encoder with shared weights and a small head reading

$$\left[\, f_{\text{pre}},\;\; f_{\text{post}},\;\; |f_{\text{pre}} - f_{\text{post}}| \,\right].$$

The absolute difference is the explicit change signal; keeping the raw features alongside it lets the head use appearance as well. Shared weights halve the parameter count and force both branches into a common feature space, which is what makes the difference term meaningful.

Model A (pretrained). Uses an ImageNet-pretrained ResNet-18 [3, 4] encoder, 11.57M parameters in total. At a 64×64 input, depth stops paying: architectures designed for 224 px inputs downsample away the little spatial resolution available, and a vision transformer would reduce a 64 px image to a handful of tokens without the data to compensate for the missing convolutional prior. ResNet-18 also keeps the comparison honest, since a much larger backbone would make any win attributable to size rather than pretraining.

Model B (from scratch). Replaces that encoder with a compact stack built for this problem: four blocks of two 3×3 convolutions with BatchNorm and ReLU followed by 2×2 max-pooling, at channel widths $32 \to 64 \to 128 \to 256$, then global average pooling. Total 1.37M parameters. Everything downstream of the encoder is identical.

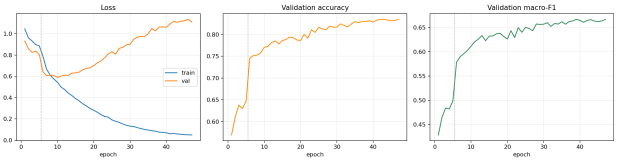


Figure 5: Model A training history. Validation loss bottoms out at epoch 11 and then rises steadily, while validation macro-F1 keeps improving to epoch 35. Model B shows the same pattern.

5.3 Training

Model A follows a linear-probe-then-fine-tune schedule [5]: phase 1 trains only the head on a frozen backbone for 5 epochs at 10^{-3}, so the randomly initialised head does not push large gradients into good weights; phase 2 unfreezes `layer3` and `layer4` for up to 45 further epochs at 10^{-4}, since early layers transfer essentially unchanged while the late semantic layers must be repurposed. Model B has no pretrained weights to protect, so the whole network trains from step one at 10^{-3} over the same 50-epoch budget.

Both use AdamW with cosine annealing, weight decay 10^{-4}, weighted cross-entropy with the class weights above, batch size 128, mixed precision, seed 42, early stopping on validation macro-F1 with patience 8, and checkpointing on the same criterion. Both are implemented in PyTorch [7] and ran locally on an NVIDIA RTX 5070 Laptop GPU; Model A stopped after 43 epochs with the best checkpoint at epoch 35; Model B stopped after 16 epochs with the best checkpoint at epoch 8, in 14.2 minutes at 53 s per epoch.

Two observations from the curves (Fig. 5) are worth recording. Unfreezing lifted Model A's validation macro-F1 from 0.499 to 0.580 in a single epoch, the clearest evidence here that a frozen ImageNet backbone is not sufficient for satellite imagery. From its epoch-11 minimum (0.588), Model A's validation loss then climbs to 1.027 by the best checkpoint and 1.130 by the end, while macro-F1 improves over the same stretch (0.633 to 0.666). Under a weighted loss on imbalanced data this is expected, since the model grows over-confident faster than it grows wrong, and it is why loss would have been the wrong early-stopping criterion. The same decoupling appears in Model B, on a differently initialised network with a different schedule, so it is a property of the objective rather than of one run.

6 Results

Table 4 gives the headline numbers. For both models `val` and `test_id` agree closely (0.666 against 0.650 macro-F1 for P, 0.597 against 0.615 for S), which matters: it shows the in-distribution figure is not a lucky validation draw, so the `test_ood` gap is a genuine domain effect rather than noise.

Table 4: Headline results. Two trained models, three evaluation sets each. P is the pretrained ResNet-18 encoder, S the from-scratch encoder. `val` and `test_id` are seen disaster types, `test_ood` is the held-out wildfire set; the drops are `test_id` to `test_ood`.

Split		Acc.	Macro-F1	AUC
`val`	P	**0.830**	**0.666**	**0.910**
	S	0.763	0.597	0.902
`test_id`	P	**0.808**	**0.650**	**0.908**
	S	0.768	0.615	0.903
`test_ood`	P	**0.774**	**0.429**	**0.814**
	S	0.630	0.377	0.790
Abs. drop	P	−0.034	−0.221	−0.094
	S	−0.138	−0.238	−0.113
Rel. drop	P	−4.2%	−33.9%	−10.4%
	S	−17.9%	−38.7%	−12.5%

6.1 Per-class behaviour

The aggregate drop is misleading, and the per-class view (Fig. 6) shows why.

For the pretrained model, destroyed scores F1 0.758 on wildfire against 0.687 in domain, at precision 0.872, and no-damage reaches precision 0.966. Despite never having been trained on wildfire imagery, the model identifies destroyed buildings under wildfire more reliably than it identifies destroyed buildings from the disaster types it was trained on. The degradation is confined entirely to the two intermediate classes, whose F1 falls from 0.480 and 0.529 in domain to 0.038 and 0.043 on wildfire. What breaks is precision, down to 0.020 and 0.024; recall does not vanish (0.284 and 0.203).

6.2 Error structure

The over-prediction is systematic rather than isolated. Against 116 true minor and 138 true major patches, the pretrained model emits 2,814 intermediate predictions and the from-scratch model 4,713, an over-prediction of 11.1× and 18.6×. That alone caps combined precision near 0.02 regardless of how good the features are. Figure 7 shows a concrete instance of the pretrained model's failure mode alongside a correct prediction from each test split.

The direction of the errors is equally consistent. For the from-scratch model, whose full confusion matrices are available (Fig. 8), 81.1% of out-of-domain errors over-estimate severity against 18.9% that underestimate, and 35.9% of truly undamaged wildfire buildings (4,324 of 12,056) are flagged as damaged. The pretrained model shows the same failure mode less severely: its no-damage recall of 0.805 on `test_ood` corresponds to a 19.5% false-alarm rate. Both confirm the prediction made from the exploratory analysis before training.

6.3 Pretrained against from-scratch

This section addresses the project's second research question directly. Both models are trained on identical patches, splits, class weights and evaluation code, and differ only in the encoder, which isolates the effect of ImageNet pretraining from every other design choice. Pretraining outperforms the from-scratch encoder on every split and every metric in Table 4, and the advantage is largest where it matters most: on `test_ood` it adds +0.143 accuracy, +0.052 macro-F1 (a 13.9% relative gain over the from-scratch model's 0.377) and +0.024 ROC-AUC. The relative macro-F1 drop from `test_id` also shrinks, from 38.7% for the from-scratch model to 33.9% for the pretrained one, so pretraining does not just raise the floor, it also makes the domain-shift degradation proportionally smaller.

Per class, the clearest single difference is destroyed: 0.758 against 0.662 out of domain, and 0.687 against 0.563 in domain. The intermediate classes are not meaningfully distinguishable between the two models, since both are far below usable precision, and at $n = 116$ and $n = 138$ the small reversals observed there fall within noise. One caveat applies to this comparison: it is not a pure pretraining ablation, since the pretrained encoder is also 8.4× larger. A third run using the same ResNet-18 architecture with random initialisation would separate the effect of pretraining from the effect of capacity; we have not run it (Section 9).

7 Discussion

Evidence for a calibration failure rather than a representation failure. Four observations support this reading of the 33.9% macro-F1 drop.

(1) destroyed scores higher out of domain than in it (0.758 against 0.687), and no-damage precision is 0.966: the two classes that actually occur under wildfire are recognised well.

(2) The errors are directionally consistent, with 81.1% over-estimating severity for the model whose full confusion matrix is available. A consistent bias is the signature of a shifted prior; a failure of the learned representation would instead be expected to produce diffuse, symmetric confusion.

(3) ROC-AUC, which does not depend on a decision threshold, holds at 0.814 against 0.908 in domain: the ranking of predictions by severity survives, and it is the decision boundary that fails.

(4) The collapse is arithmetically forced by the class-frequency shift. minor and major make up 10.5% and 8.6% of training patches but only 0.79% and 0.94% of wildfire patches. The pretrained model emits 2,814 intermediate predictions against 254 true cases, which caps combined precision near 0.02 regardless of feature quality, and matches the 0.020 and 0.024 actually measured.

This distinction has direct operational consequences. A representation failure would mean the approach cannot be deployed on a new disaster type at all. A calibration failure means the ranking of predictions is usable and only the decision rule needs adjusting, a considerably more tractable problem with an established remedy in prior correction [6], which rescales

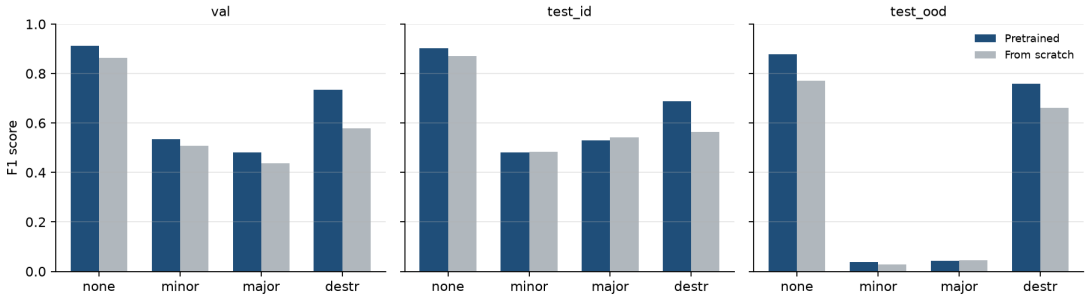


Figure 6: Per-class F1 for both models on the three evaluation sets. In domain the two are close on the intermediate classes and pretraining mostly buys destroyed; out of domain both collapse on minor and major, and the pretrained advantage on destroyed widens.

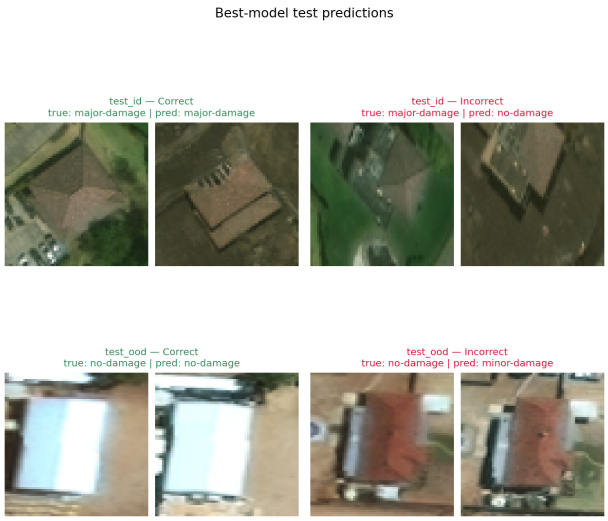


Figure 7: Pretrained model: one correct and one incorrect prediction on `test_id` and on `test_ood`, pre-disaster crop left and post-disaster crop right in each pair. The `test_ood` error is a truly undamaged building predicted minor-damage, the over-flagging failure mode quantified above.

class posteriors by the ratio of target to source priors without retraining. A model of this kind deployed on an unlabelled new disaster type would reliably separate destroyed from intact structures while substantially over-reporting intermediate damage. For triage, i.e. prioritising where responders look first, that failure mode is comparatively benign; for damage quantification, and particularly for aid allocation, it is not.

8 Impact and recovery analysis

8.1 Data and method

To ask whether damage falls unequally across communities, we take the per-building WGS84 centroids in the xBD labels, restrict to the ten US events for which census geography is available, and spatially join them to 2020 TIGER census block groups. Each building contributes an ordinal damage score (no-damage = 0 to destroyed = 3); the unit of analysis is the block-group mean. Block groups with fewer than 20 buildings are dropped. That gives 185,045 buildings in 308 block groups, joined to American Community Survey covariates for 2017 [8].

Two data-quality problems had to be handled first, and both are reported rather than absorbed silently.

Table 5: Standardised OLS coefficients on block-group mean damage score, $n = 291$. Each column adds controls to the one before it: socioeconomic only, plus built environment, plus event fixed effects.

	naive	+ built	+ FE
Median income	$+0.211^{**}$	$+0.225^{***}$	$+0.057$
Median home value	-0.150^{**}	-0.139^{*}	-0.039
Renter share	$+0.015$	$+0.032$	-0.016
Building density		-0.085^{*}	-0.103^{**}
Building age		-0.102^{*}	-0.096^{*}
R^2	0.045	0.072	0.225
adjusted R^2	0.035	0.055	0.186

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Census sentinel values. The ACS extract encodes missing values as -666666666 rather than as nulls, and 17 of the 308 block groups carry one in median income, home value or year built, which if left in place makes those coefficients meaningless. We drop them, leaving 291 block groups.

Two unusable columns. The poverty rate is empty for every block group, and the hazard exposure proxy is identically zero because no fire-perimeter, storm-track or coastline layer was ever joined. Both are dropped rather than reported as null effects.

The exposure control is therefore the built environment only (building density and median building age), plus, in the strongest specification, event fixed effects. Event identity is the sharpest exposure control this data supports: comparing two block groups inside the same disaster holds the hazard roughly constant.

8.2 Results

Table 5 summarises the result. Regressed on socioeconomic variables alone, median income carries a significant positive coefficient ($+0.211$, $p = 0.002$), and adding built-environment controls strengthens it fur-

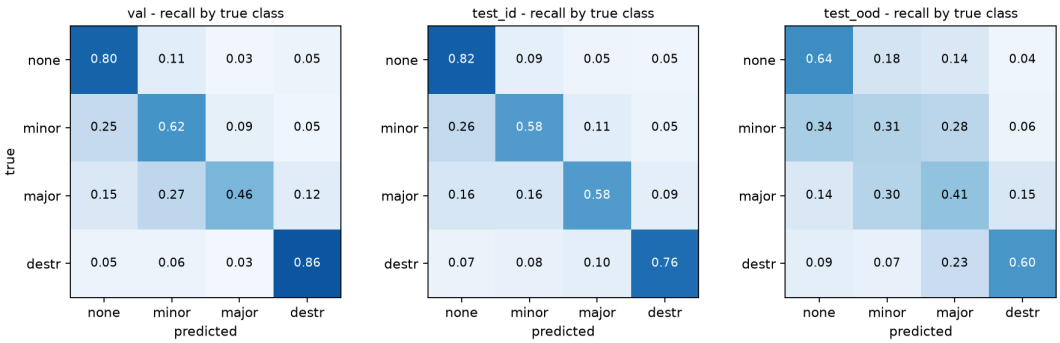


Figure 8: Confusion matrices for the from-scratch model, normalised per true class (recall). The `test_ood` panel shows probability mass pushed off the diagonal in one direction, toward greater severity, rather than scattered.

ther (+0.225, $p = 0.0009$). Taken at face value, this would indicate that richer block groups suffered more damage.

Adding event fixed effects destroys the effect. The income coefficient falls to +0.057 and is no longer distinguishable from zero ($p = 0.44$), while R^2 more than triples, from 0.072 to 0.225. Almost all of the apparent socioeconomic signal was between-event variation: `hurricane-harvey` block groups have both high income ($98k median) and high damage (mean score 0.88), while `santa-rosa-wildfire` and `moore-tornado` have moderate income and low damage.

Figure 9 shows the same thing without a model. Pooled across events the relationship is U-shaped, not monotone: mean damage score is 0.58 in the poorest income quintile, falls to 0.33 in the middle, and rises to 0.61 in the richest. That is what a mixture of two exposure stories looks like, low-income tracts in tornado paths and high-income tracts in coastal and wildland-urban-interface zones, rather than a single inequality gradient. Computing income terciles within each event, over the six events with at least 15 block groups, flattens it to 0.50 / 0.43 / 0.50, and the per-event direction is inconsistent in sign: damage rises with income across `hurricane-harvey` (0.77 to 1.01) and falls with it across `joplin-tornado` (0.90 to 0.41).

What survives event fixed effects is the built environment: building density (−0.103, $p = 0.006$) and median building age (−0.096, $p = 0.012$) both remain significant and barely move (Fig. 10). Denser and older block groups record less damage on average, which runs against the usual intuition that older housing is more vulnerable, and is most plausibly a settlement-pattern effect: dense, older cores sit away from the coastal and interface edges where these particular hazards concentrate.

8.3 Pricing damage, and checking the price against real records

An ordinal damage class says nothing about the size of the loss it represents, and the project specification asks about recovery as well as damage. To get at that, every building's damage class is converted to a dollar figure using the HAZUS-MH per-damage-state loss ratios [10] (0% for no-damage, 5% minor-damage, 30% major-damage, 100% destroyed) applied to its block group's median home value. The rule is not disaster-specific: a destroyed house costs its replacement value whichever hazard destroyed it. Over the 108,491 buildings inside the 291 analysis block groups this prices $6.76B of modelled loss, with wildfire alone accounting for 70% of it (Californian home values are high and fire tends to destroy outright), hurricane 22%, tornado 7%, and the single flood event under 1%, so the dollar layer is not simply a restatement of flood risk.

That price was checked against three independent, real-world records rather than taken on faith. Against 6,525 National Flood Insurance Program settlements [11] inside the same block groups, the modelled price lands at 0.32–0.85× what the adjuster assessed on site: the right order of magnitude, and consistently conservative. Against 9,395 CAL FIRE post-fire structure inspections [12] for the three wildfire events, inspectors measured an average loss of 91.4% of a structure where the assumed ratios give 92.7%, within 1.3 points. And against 133,240 FEMA household loss verifications [11] spanning all ten events, the modelled price runs well above FEMA's own figure, which is expected: FEMA's verified loss records the damage the aid programme will pay for, a floor on the rebuild cost rather than an estimate of it.

8.4 Recovery capacity is where the real inequality is

Figure 11 makes a point the regression above cannot. Physical damage is close to flat across income (10.6% of building value lost in the poorest quintile against 12.1% in the richest, non-monotone), but the identical damage map priced in dollars is steep, rising from $13k to $89k in modelled loss per building, a ∼7× gradient, simply because a home in a richer block group costs more to begin with.

Who actually pays that bill tracks income even more sharply (Fig. 12). Restricting to the four water events, where the National Flood Insurance Program publishes a claim for every settled property, claims filed per damaged building rise steeply with income (Spearman $\rho = 0.62$): the poorest income quartile files a claim for 5% of its damaged buildings against 51% in the richest. Splitting the repair bill three ways tells the

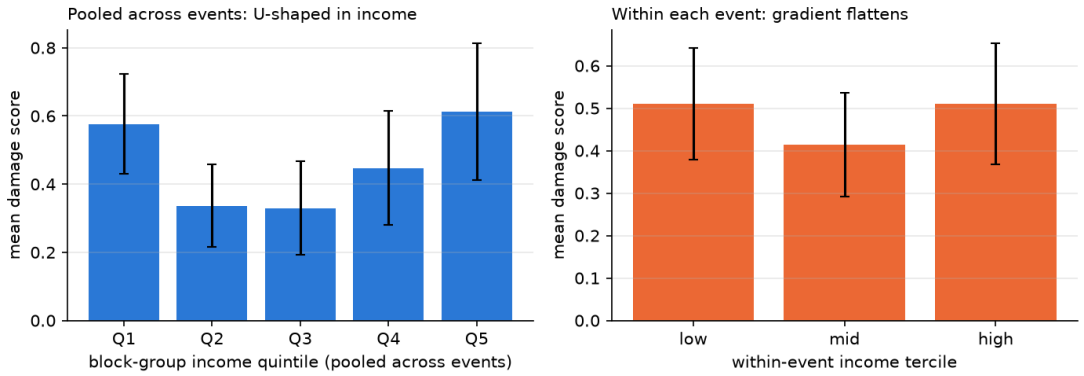


Figure 9: Left: pooling all ten events, mean damage score is U-shaped in income, highest in the poorest and the richest quintiles and lowest in the middle. Right: once income terciles are computed *within* each event, the gradient flattens and the 95% intervals overlap completely.

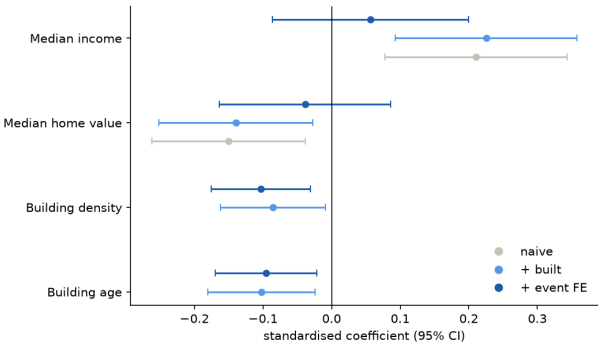


Figure 10: Standardised coefficients with 95% intervals across the three specifications. The two socioeconomic effects shrink toward zero when event fixed effects enter; the two built-environment effects do not.

same story: insurance covers 23% of the poorest quartile's modelled loss and FEMA grants another 16%, leaving 61% covered by nothing, while in the richest quartile insurance alone covers 65% and only 34% goes uncovered.

FEMA's own applicant records extend this picture to every disaster type [11]. Among 115,463 owner-occupants who applied for federal aid, the insured share rises from 38% in the poorest household-income band to 95% in the richest, and the gradient holds inside every disaster type. This also corrects a flood-only impression: wildfire households are the best insured of the four (75%), and flood households the worst (60%), because a standard homeowner policy covers fire and wind while flood requires a separate NFIP policy.

8.5 Auditing the predicted damage map

Every result up to this point uses xBD ground-truth labels, by design (Section 8.6 explains why). The last question this data can answer directly is what changes if a relief programme used the Part 2 models' predictions instead of verified labels. Joining the two models' output on their held-out splits to the same census geography gives 12,438 buildings across seven events (three hurricanes, one flood, three wildfires) and 122 block groups; tornado is absent from the audit for an unrelated reason, since all three tornado events sit entirely inside the Part 2 training split and so have no held-out predicted map.

We split a fixed $1B relief budget by income tercile twice, once on the true damage map and once on the pretrained model's predicted map, under two allocation rules (an equal payment per damaged building, and a payment proportional to modelled dollar loss), and measure how many dollars land in a different income tercile than the labels would send them. On the seen disaster types the predicted map misdirects $30–78M of every $1B; on wildfire, the type the model never trained on, it is $112–200M (Fig. 13), and the direction is systematic rather than random: consistent with the intermediate-class over-prediction documented in Section 6, the model invents damage in mid-income block groups and withdraws funding from the high-income block groups where the wildfires actually destroyed the housing. The predicted map also over-states the total dollar loss on this sample by 19%. This is not a flaw specific to the pretrained model: the from-scratch model's audit numbers are of the same order ($37–81M seen, $137–179M wildfire).

8.6 What this means for deployment

Three cautions follow from this analysis, and they compound. First, the naive specification would have supported a confident but incorrect policy claim: a pipeline that regressed damage on income and reported the coefficient would have published a significant positive income effect that does not survive the most basic exposure control available. The confounder here is not subtle; it is simply which disaster struck which place.

Second, the recovery-capacity analysis in Section 8.4 shows an inequality the damage regression cannot see: physical damage is close to flat across income, but dollar loss and the capacity to cover it are not. A relief rule keyed to dollar damage, or to insurance status, is regressive before any model is involved.

Third, Section 6 showed that the model's errors are not random: it over-flags intermediate damage at a rate that depends on the disaster type, and Section 8.5

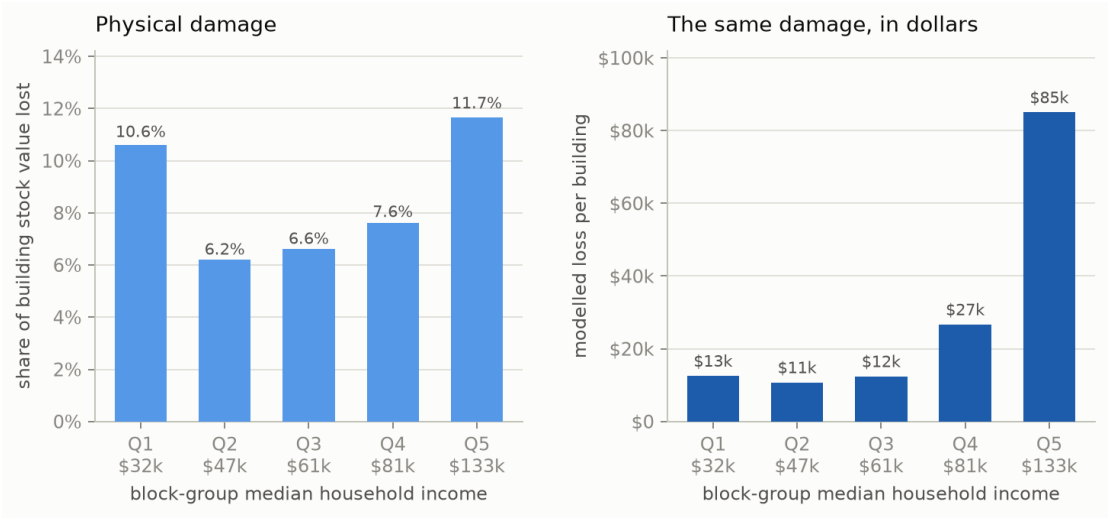


Figure 11: Left: share of building value lost, by block-group income quintile. Right: the identical damage map, priced in dollars per building. Physical damage is roughly flat; the dollar figure is not.

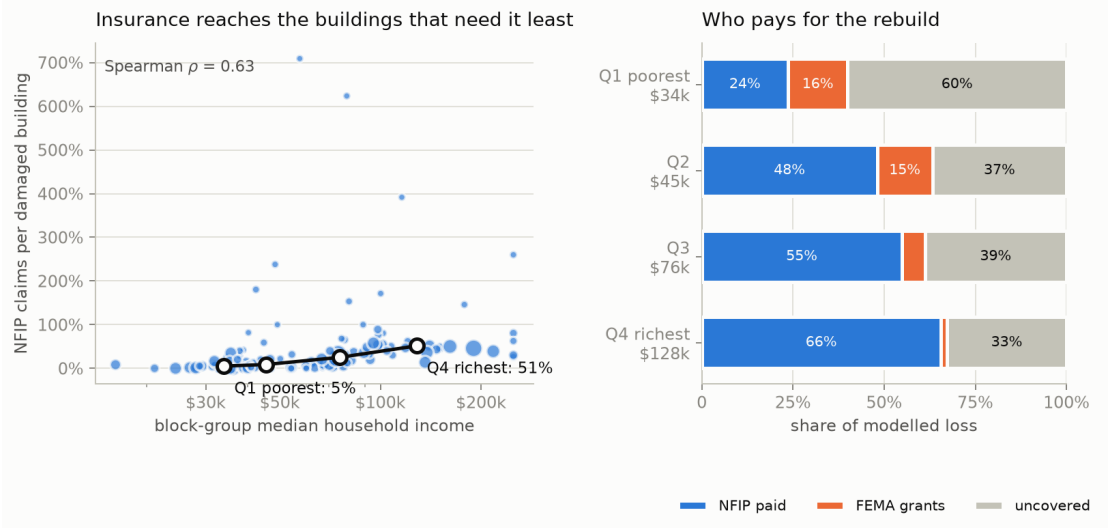


Figure 12: Left: NFIP claims filed per damaged building against block-group income, the four water events. Right: the repair bill split into what insurance paid, what FEMA paid, and what nobody paid, by income quartile.

shows what that costs in practice, moving real relief money along income lines and moving more of it on the one disaster type the model has never seen. Rapid automated damage maps are useful for triage; using them as evidence for aid allocation without verified labels risks more than generic error, it risks a specific, income-correlated error, in the direction the model already over-predicts.

9 Limitations

- **Thin out-of-domain support.** minor and major rest on 116 and 138 true examples in `test_ood`. These two classes drive the macro-F1 story, so their per-class figures carry wide uncertainty.
- **Annotator selection.** Every number is measured on buildings annotators were able to assess, since un-classified polygons are excluded, so true zero-shot difficulty is somewhat higher than reported.
- **Confounded geometry and partial checks.** The sun-elevation shift is inseparable from event identity with only 19 events, the duplicate test covers 400 of 11,034 scenes, and model results use a stratified sample of 1,698 scenes.
- **Not a pure pretraining ablation**, since the encoders differ in initialisation and in size (11.57M against 1.37M parameters); and model selection uses `val`, which is in-distribution, because selecting on the out-of-domain set would leak the test set.
- **Ecological inference in Part 3.** Block-group averages cannot support claims about individual households, no true hazard-exposure layer was available, and ten events is a small basis for fixed effects.
- **Recovery-analysis assumptions.** One median home value stands in for every building in a block group, and four HAZUS loss ratios stand in for a continuous damage distribution. Tornado has no inspection dataset comparable to CAL FIRE's, so its loss ratios rest on the HAZUS assumption alone. Insurance take-up from FEMA records describes

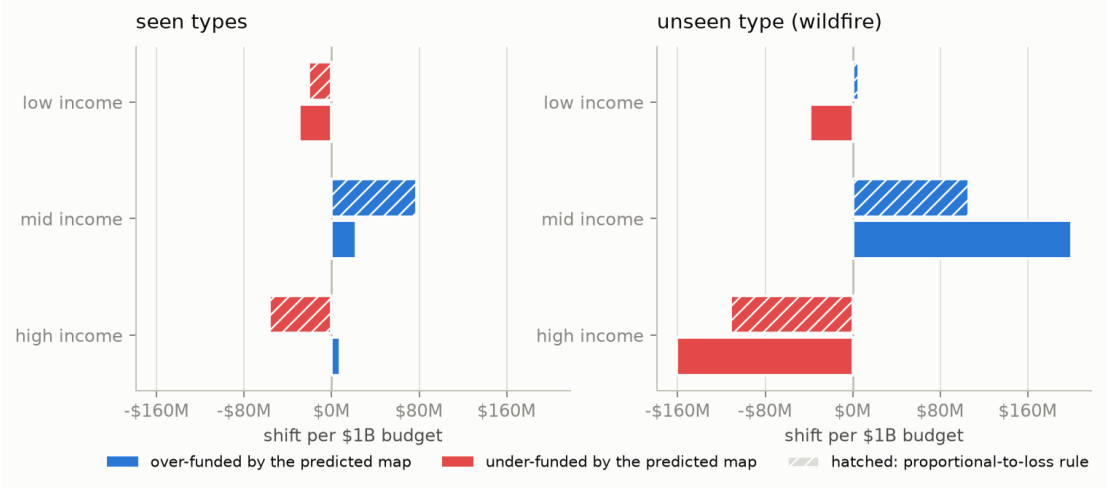


Figure 13: Dollars per $1B relief budget that land in a different income tercile under the predicted map than under the true damage map, pretrained model. Blue over-funds a tercile, red under-funds it. The wildfire panel is both larger and directionally consistent, unlike the seen-type panel.

owner-occupants who applied for aid, not the general population, and the predicted-damage audit covers seven of the ten events because the three tornado events never appear in a held-out split.

10 Conclusions and next steps

This project set out to answer two questions: whether building-damage classification transfers to a disaster type withheld from training, and whether ImageNet pretraining improves that transfer relative to a comparable encoder trained from scratch. On the first question, a siamese change-detection CNN trained on 14 non-fire disasters and evaluated zero-shot on five wildfire events loses a third of its macro-F1, but the loss is confined to the two damage classes that barely exist under wildfire, while destroyed-versus-intact discrimination improves out of domain. On the second, ImageNet pretraining outperforms the from-scratch encoder on every split and every metric, with the largest margin precisely on the wildfire set. A supplementary impact analysis finds that raw damage is not distributed along an income gradient once the disaster event itself is controlled for, but that finding does not extend to recovery: priced in dollars, and measured against real insurance and federal-aid records, the same damage falls on households with sharply unequal capacity to pay for it, and auditing the Part 2 models' own predictions shows that an automated relief rule would move real money along income lines, more so on the disaster type the models never trained on. Outstanding work, in priority order:

1. **Test the calibration hypothesis directly.** Apply prior correction [6] to the out-of-domain predictions of both models and measure how much of the macro-F1 gap closes without retraining. If most of it closes, that is the strongest result the project can report.
2. **Few-shot ablation**: add 1 to 5% of wildfire scenes to training and re-measure the gap.
3. **Randomly initialised ResNet-18** as a third run, to separate pretraining from capacity.
4. **Real hazard-exposure layers** for Part 3, in place of the event-fixed-effects proxy, and a tornado inspection dataset comparable to CAL FIRE's to remove the last HAZUS-only loss ratio.

Reproducibility. Every figure, table and number in this report is generated from a CSV artefact written by the notebooks and scripts in the repository; nothing is hand-computed. Seed 42 throughout. Environment setup, data sources, and a map from each notebook to the results and figures it produces are documented in the repository `README.md`.

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